Codon–backbone sensitivity analysis: comparison of statistics

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Summary

After correcting the per-position aggregation (key = (unp_id, unp_idx, secondary structure); removal of gross conformational outliers — §0, shown to be inconsequential), we re-examined the synonymous-codon backbone signal under **five two-sample statistics**, reporting for each (i) the FDR-rejected codon pairs and (ii) an adversarial **breakdown-k** sensitivity analysis, calibrated against a spurious-but-significant reference. Two pairs — **A-GCG:A-GCT (TURN)** and **L-CTC:L-TTG (HELIX)** — are significant and breakdown-robust across statistics; the remaining rejections are statistic-specific. We recommend the **unbiased MMD²** statistic as the primary one, because it removes the KDE-L1 sample-size bias by construction and (with a bounded kernel) is outlier-robust.

0. Data cleanup: aggregation key and conformational outliers

Before any statistic, two data-quality issues in the per-position aggregation were fixed. Both concern how the dihedral angles of a residue are pooled across the PDB structures in which it appears.

(a) Aggregation key. The original pipeline aggregated angles per **(unp_id, unp_idx)** — one centroid per UniProt sequence position, pooling *all* structures of that residue regardless of conformation. But a residue can adopt different backbone states (and DSSP secondary-structure assignments) in different structures, so pooling across structure can average genuinely distinct conformations into a meaningless centroid. We now aggregate per **(unp_id, unp_idx, secondary structure)**, keeping distinct conformational states separate.

(b) Conformational outliers. Within each aggregation group of ≥3 structures, a structure whose (ϕ,ψ) lies more than **60°** (flat-torus distance) from the group's robust centre — i.e. in a clearly different Ramachandran basin — is removed before averaging, and the centroid recomputed. Applied uniformly across the whole dataset.

Size of the phenomenon.

effect	count	fraction
positions split across >1 SS class (key fix)	1,569	1.5% of ~101k
outlier structures removed (≥3-structure groups)	365	0.14% of per-structure records
positions affected by outlier removal	200	0.2%

Removed outliers sit a median of **156°** from their group consensus (genuinely different basins), and are concentrated in flexible regions — by SS: HELIX 7, SHEET 27, TURN 129, OTHER 202 (rare in well-ordered helices, common in turns/loops).

Why this is inconsequential for our conclusions. We re-ran the *entire* analysis — both KDE-L1 and torus arms, all controls, all breakdown-k — on the cleaned dataset. **Every significance decision, breakdown-k, and control outcome is reproduced unchanged.** The reasons:

- The base reconstruction matches the original aggregation to a **median of 0.000°** at unaffected positions; only 200 of ~101k positions move at all.
- Of the 365 removed structures, **exactly one** falls on a kill-set (signal-carrying) residue — 2NOO in P33590:421, whose aggregated spread drops from σ = 26° to < 8° once removed.

- The signal-carrying residues were already well-determined (median circular std ≈ 1.8°, all in allowed Ramachandran regions), so removing outliers — which lie overwhelmingly in flexible regions away from the rejected pairs — leaves them untouched.

The only measurable change anywhere is A-GCG:A-GCT's torus statistic shifting 0.466 -> 0.460 (still far below threshold, still never breaking). The phenomenon is real and worth fixing for correctness, but it does not alter any conclusion in this report. All results below use the cleaned, (unp_id, unp_idx, SS)-aggregated dataset.

1. Statistics compared

short name	statistic	bandwidth	notes
KDE-fix	KDE-L1 on torus	fixed σ = 10°	original 2022 statistic
KDE-CV	KDE-L1, per-(codon,SS) σ	LOO-CV (double-slab)	revision's principled bandwidth
RBF-MMD	MMD² (biased, V-stat)	Gaussian, σ = 10°	bounded-influence
MMD²-unbiased	MMD² (U-stat)	Gaussian, σ = 10°	**size-unbiased + bounded**
torus-W2	projected Wasserstein	4 fixed geodesics	González-Delgado test

All on the bit-exact reproduced dataset (101,419 positions); permutation tests K=5000; FDR = 0.05, BH per secondary-structure class; "noself" filter; AA+SS randomized control.

2. Rejected pairs (FDR = 0.05)

Candidate set = the union of pairs rejected by any statistic (6 pairs).

statistic	# rejected	rejected pairs
KDE-fix	4	L-CTC:L-TTG, L-CTC:L-CTG (HELIX); A-GCG:A-GCT, P-CCC:P-CCG (TURN)
KDE-CV	2	L-CTC:L-TTG (HELIX); A-GCG:A-GCT (TURN)
RBF-MMD	2	L-CTC:L-TTG (HELIX); A-GCG:A-GCT (TURN)
MMD²-unbiased	2	L-CTC:L-TTG (HELIX); A-GCG:A-GCT (TURN)
torus-W2	5	L-CTC:L-TTG, L-CTC:L-CTG, L-CTC:L-CTT, R-AGG:R-CGA (HELIX); A-GCG:A-GCT (TURN)

(MMD significance assessed at the published per-SS BH levels — HELIX 0.00115, TURN 0.00057 — for a common footing; a statistic-specific MMD-BH over all 87 pairs is a later refinement. P-CCC:P-CCG is a borderline miss under MMD, p≈0.0006–0.001.)

3. Sensitivity — adversarial breakdown-k

Greedy removal of the most influential observations (re-ranked each step) until the pair leaves FDR significance. "robust" = never broke within cap; "boundary" = exactly at the FDR boundary.

SS	pair	KDE-fix	KDE-CV	RBF-MMD	**MMD²-unb.**	torus-W2
HELIX	**L-CTC:L-TTG**	k=8	boundary	k=5	**k=5**	robust
HELIX	L-CTC:L-CTG	k=1	n.s.	n.s.	n.s.	k=20

SS	pair	KDE-fix	KDE-CV	RBF-MMD	**MMD²-unb.**	torus-W2
HELIX	L-CTC:L-CTT	n.s.	n.s.	n.s.	n.s.	k=20
HELIX	R-AGG:R-CGA	n.s.	n.s.	n.s.	n.s.	k=12
TURN	**A-GCG:A-GCT**	k=8	k=2	k=8	**k=8**	robust
TURN	P-CCC:P-CCG	k=8	n.s.	n.s.	n.s.	n.s.

Caveat: breakdown-k depends on the BH threshold, which depends on how many rejections occur in that SS class (e.g. KDE-CV's k=2 vs KDE-fix's k=8 for A-GCG:A-GCT is partly the tighter CV threshold, 0.00057 vs 0.00115). breakdown-k should be read together with the p(k) trajectory, not as a single absolute number.

4. Sample-size sensitivity (and the fix)

KDE-L1 is a **biased** estimator of the density distance — smaller samples give larger estimated distances:

statistic	Spearman(min n, distance) over 316 pairs
KDE-L1	**−0.64** ($p = 2\times10^{-37}$)
unbiased MMD²	**+0.04** ($p = 0.46$, null)

The unbiased MMD² (U-statistic, mean-zero under H_0 at any n) **removes the bias by construction** — no subsampling needed, effect sizes comparable across pairs. With a bounded Gaussian kernel it is simultaneously **outlier-robust**, and it uses a single kernel width (no per-pair CV fragility). Hence the recommendation to adopt it as the primary statistic.

5. A reference for breakdown-k

control breakdown-k = 0 is tautological (no AA+SS control pair is significant). We instead manufactured **spurious-but-significant** pairs: split one amino acid's residues in one SS into two pseudo-codon groups by a random circular-aware direction in (ϕ,ψ) + tunable noise, matched to each real pair's n and p, under unbiased MMD².

matched to	real breakdown-k	spurious breakdown-k
A-GCG:A-GCT (p≈floor)	8	≥21 (all reps; never broke)
L-CTC:L-TTG (p≈0.0008)	5	median 20 (bimodal: 0 or ≥20)

Interpretation: a *genuinely distributed* difference at the same significance is **more** breakdown-robust (k≥20) than the real pairs (5–8). So at matched p, breakdown-k does carry information beyond the p-value — and the real codon differences are **real but comparatively concentrated** (carried by fewer points than a fully distributed effect). The spurious construction never yields a *fragile* significant rejection (it is bimodal: non-significant, or robustly significant), and the AA+SS controls give 0/30 — so the real pairs are not competing with noise.

6. Conclusions

1. **A-GCG:A-GCT** and **L-CTC:L-TTG** are significant under (essentially) all five statistics and survive 5–8 adversarial deletions — the solid claims.
2. **L-CTC:L-CTG, L-CTC:L-CTT, R-AGG:R-CGA** are torus-only; **P-CCC:P-CCG** is KDE-fix-only. These four are statistic-dependent and should be reported with that caveat.
3. **Adopt unbiased MMD²** as the primary statistic: it removes the sample-size bias, is outlier-robust, and avoids bandwidth fragility — addressing the main methodological critiques in one move.
4. Report breakdown-k **with p(k) trajectories and the spurious reference**, not as a single absolute number, because it is threshold- and significance-dependent.

Part II — Biological significance

The statistics above establish that the codon–backbone signal is **real**. Part II asks whether it has a **biological mechanism**, probing the four axes a reviewer would raise. Throughout, conclusions are anchored on the two pairs that are robust across all statistics — **A-GCG:A-GCT (TURN)** and **L-CTC:L-TTG (HELIX)** — and the size-unbiased **unbiased MMD²** statistic is used wherever a distributional distance is needed, so that none of the biological conclusions inherit the KDE-L1 sample-size bias.

7. Structural quality — are the residues well-determined?

Methodology. B-factors are not comparable across structures, so we z-normalise B within each PDB structure and average per aggregation position (B_z). We compare B_z between the two codons with a Mann–Whitney test (a rank test on a scalar — not a density distance, hence immune to the sample-size bias), and check each kill-set residue against the core-allowed Ramachandran regions.

Result. Kill-set residues are **rigid** (median B_z ≈ −0.34, well below the proteome mean), the two codons are **equally flexible** (MWU n.s.), and **100%** of kill-set residues lie in allowed Ramachandran regions. The signal therefore sits on well-ordered atoms — **not a disorder or mis-modelling artifact**. (37/63 kill-set residues are single-PDB; these are equally rigid and allowed, but cannot be cross-validated across structures — a residual caveat.)

8. Translation speed — does the shift track decoding rate?

Data. Experimental E. coli codon speeds from **Chevance et al. 2014** (his-leader operon expression; high = slow), digitised from their Fig. 3 and verified against the published figure. Speeds are compared as **ratios / log-ratios**, not differences, since the units are arbitrary.

Methodology. Across all synonymous codon pairs (within AA × SS) we correlate |log speed-ratio| with the structural divergence measured by **unbiased MMD²**. This statistic is essential here: the KDE-L1 density distance is sample-size-biased (Spearman(min n, KDE-L1) = −0.64), and that bias is exactly what produced the spurious $r^2 = 0.6$ speed<->distance trend reported earlier (rare = slow = small-n = inflated distance). Unbiased MMD² has Spearman(min n, MMD²) = +0.04 — no bias.

Result. **No relationship** — Spearman(|log speed-ratio|, MMD²) = +0.04 ($p = 0.44$). The simple co-translational-**speed** mechanism is **not supported**; the earlier speed correlation was a sample-size artifact.

9. Per-amino-acid directionality — is any single residue special?

Methodology. A pooled test can mask an amino-acid-specific effect. For each amino acid we correlate codon speed with the **circular mean** (φ,ψ) of its residues, SS-centred (so secondary structure is controlled), with a permutation test over the codon->speed labels. Circular means are used (not density distances) because they converge cleanly with n.

Result. Only **proline in turns** shows a clean monotonic trend — slower proline codon -> higher ψ (toward PPII), CCG < CCA < CCU < CCC in both speed and ψ, spanning ~24° (codon-mean Spearman = 1.0). But it is **underpowered** (proline has only 4 codons -> permutation floor $p \approx 0.08$–0.17), **confounded with 3rd-nucleotide identity**, and sits on a pair (P-CCC:P-CCG) that is **not robust across statistics** (KDE-fix-only). It is a **speculative, hypothesis-generating** lead, not an established effect.

10. Expression confounder (Cope–Gilchrist)

Data. A non-circular expression axis — **PaxDb** integrated protein abundance for E. coli K-12 — joined to each position's gene via UniProt -> locus tag (72% of proteins mapped).

Methodology. Positions are split into LOW/MID/HIGH gene-abundance tertiles. Within each stratum (where expression is ~constant) we recompute each pair's **unbiased MMD²** and its permutation p. Because MMD² is size-unbiased, the cross-stratum effect **magnitudes are directly comparable**.

Result. For the robust pairs the effect **magnitude is concentrated in high-expression genes** (MMD² ≈ 6–20× larger in HIGH than LOW/MID: L-CTC:L-TTG 0.006/0.002/**0.034**; A-GCG:A-GCT 0.004/−0.002/**0.058**). Two implications: (i) the effect is **present within** the high-expression stratum, so it is **not an expression artifact** (the simplest Cope–Gilchrist account is rejected); (ii) but it is **expression-modulated** — strongest where translational flux and codon selection are both highest, which this observational design cannot separate. Per-stratum *significance* is weaker than the magnitude implies, because the abundance-mapped subset (76%), split into thirds, leaves ~¼ of the data per stratum.

11. Biological synthesis

axis	finding
structural order	on rigid, well-ordered, allowed-region residues — not a disorder artifact
translation speed	no relationship (size-corrected) — not a clean speed mechanism
per-AA directionality	only proline-in-turns suggestive — speculative (underpowered, confounded)
expression	magnitude concentrated in high-expression genes; not an expression artifact, but expression-modulated
concentration	breakdown-k 5–8 vs distributed-fake ≥20 — real but carried by a sub-population

Defensible position: the codon-conditioned backbone signal is **real, specific, on well-determined residues, and not explained by expression** — yet it does **not** behave like a clean translation-speed effect once sample size is controlled, and it is strongest in highly-translated genes. The mechanism remains **open**, with a concentrated, expression-linked, co-translational *flavour* but no single mechanism established. This is the cautious framing the Brief Communication adopts, now supported on every axis a referee is likely to probe.